Concepts of Programming Languages

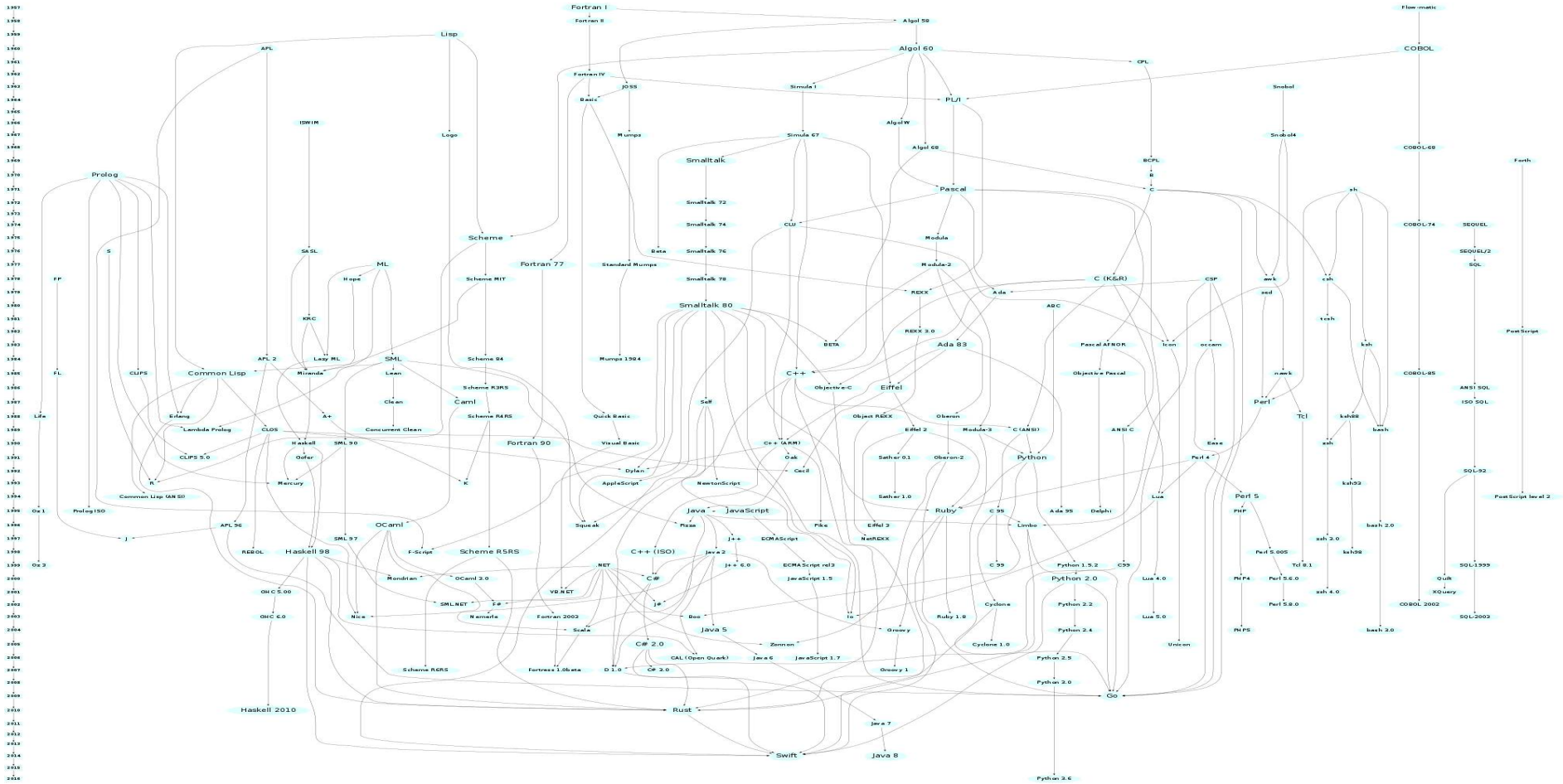
Overview

4 October 2018

Johannes Weigend
Weigend AM

Evolution of Programming Languages

./img/01-languages.png (./img/01-languages.png)



Types of Programming Languages

- **Imperative**: Assembler as root (Operations and Control Structures)
- **Declarative:** Domain Specific Languages (HTML, XML, YAML, ...) => Limited in scope. Some are not Turing Complete.
- **Procedural** (Control Structures and Procedures): Fortran / Algol / Pascal / **C**
- **Object Oriented** (Classes and Objects): Smalltalk / Objective C / C++ / Eifel / **Java** / C#
- **Functional** (Lambdas + Closures): Haskell, Erlang, F#
- **Logic**: Prolog
- **Best of Breed**: Kotlin, Scala, Rust, JS, **Go**

More Languages (https://en.wikipedia.org/wiki/List_of_programming_languages_by_type)

Areas for Programming Languages

- Application Programming (small / large scale, web, central, cloud)
- Systems Programming
- Network Programming / Distributed Systems
- Realtime Systems
- User Interface Programming
- Embedded Systems

Orthogonal Classifications

- Static vs. Dynamic Typed
- Compiled vs. Interpreted
- Sequential vs. Parallel
- Static Linked vs. Dynamic Linked
- Safe vs. Unsafe (Crash a process or even the system)
- Simple vs. Complex
- Platform Independent Bytecode vs. Platform Dependent Assemblercode
- Platform Dependent Compiler vs. Cross Compiler

Some Thoughts

- There are many solutions for a given problem
- The "best" language is nothing without a proper ecosystem (Libraries, Examples, Community)
- There are straight forward choices for a given platform (Mac: Swift + Objective-C, Linux: C + Python, Android: Java/Kotlin)
- Don't expect "Magic" from programming Languages
- Clean code and good programming is orthogonal to the language
- Learn more than one programming language at a professional level

Thank you

Johannes Weigend
Weigend AM
johannes.weigend@weigend.de (mailto:johannes.weigend@weigend.de)
http://www.weigend.de (http://www.weigend.de)
@johannesweigend (http://twitter.com/johannesweigend)